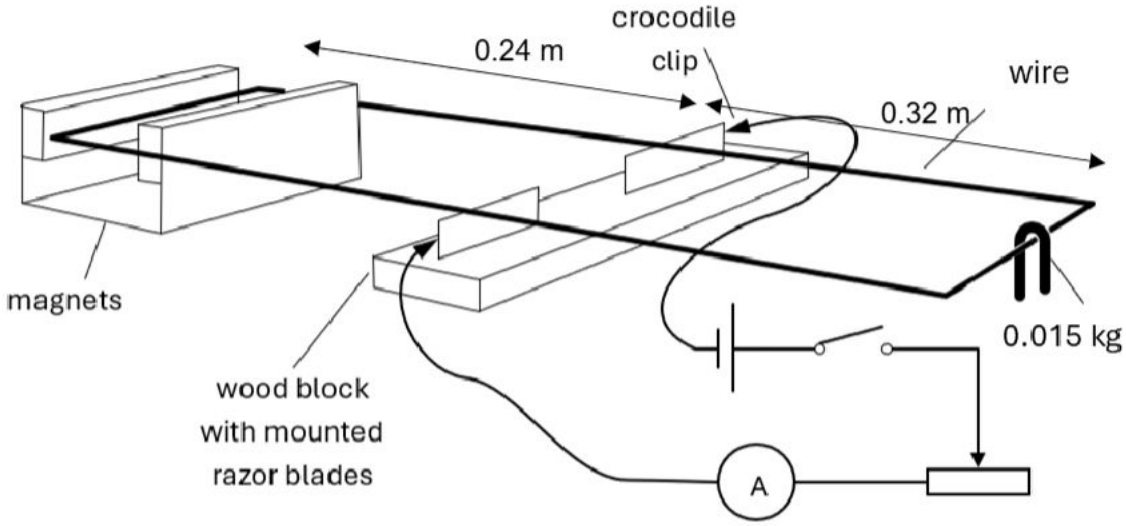
22 A laboratory current balance consists of a horizontal copper wire supported on two metal supports. A section of wire of length 0.18 m lies perpendicular to a uniform magnetic field produced by a magnet. When a current flows in the wire, a magnetic force due to the magnetic field acts on it.



The magnetic force acts at a perpendicular distance of 0.24 m from the pivot. A hanging mass 0.015 kg, placed 0.32 m from the pivot, balances the wire when the switch is turned on and the current is 1.5 A.

What is the magnetic flux density produced by the magnet?

A 0.18 T **B** 0.48 T **C** 0.73 T **D** 0.95 T

Ans: C

Moment due to magnetic force:
$\tau_B = BIL(0.240)$

Moment due to hanging mass:
$\tau_m = mg(0.320)$

Apply Principle of moments:

$$\tau_B = \tau_m$$
$$BIL(0.240) = mg(0.320)$$
$$B = \frac{(0.0150)(9.81)(0.320)}{(1.50)(0.180)(0.240)}$$
$$= 0.727 \text{ T}$$